STEPHEN NYARANGA

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stephen-nyaranga.github.io | linkedin.com/in/stephennyaranga | ORCID 0009-0001-2577-6994 | Kenyan

Climate scientist and geophysicist. My doctoral research combines Sentinel-1 SAR and deep learning for flood extent mapping, flood-depth estimation and urban resilience in Rwanda, as part of the CLARITY-Africa programme.

EDUCATION

PhD in Climate Science -- CLARITY-Africa Fellowship *Oct 2025 -- Present*
AIMS-RIC Doctoral Training Program, Kigali | Host university: Jomo Kenyatta University of Agriculture and Technology (JKUAT), Dept. of Geomatic Engineering & GIS | Imperial College London

- Full 36-month CLARITY-Africa scholarship awarded by AIMS-RIC (see Research Affiliations).
- Research: Integrating Deep Learning and SAR Analysis for Flood Extent Mapping, Depth Estimation, and Urban Resilience in Musanze and Kigali, Rwanda.
- Supervisors: Prof. Dr. Mouhamadou Bamba Sylla (AIMS-RIC, main); Dr. Ben Howard and Prof. Dr. Wouter Buytaert (Imperial College London); Dr. Mercy Mwaniki (JKUAT, host).

MSc Geophysics (Pursued) *Jan 2023 -- Dec 2024*
ICTP-East African Institute for Fundamental Research (ICTP-EAIFR), Kigali, Rwanda

- Thesis: Time Series InSAR Analysis over the Olkaria Geothermal Power Plant in Kenya: Insights From a 2020--2024 Sentinel-1 InSAR Survey.
- Coursework: seismology, geodesy, InSAR, computational geodynamics, earthquake mechanics, atmospheric physics, GPS geodesy, physics of volcanoes.

MSc Mathematical Sciences -- Climate Science / Data Science *Oct 2021 -- Jul 2022*
African Institute for Mathematical Sciences (AIMS-Senegal), Mbour, Senegal

- Graduated with Distinction. Mastercard Foundation Scholar.
- Dissertation: Atmospheric Rivers and Surface Melting over the Pine Island Glacier in 1983 and 1998.

BSc Education Science (Physics / Mathematics) *Aug 2016 -- Mar 2021*
Karatina University, Nyeri, Kenya

- First Class Honours. Registered with the Teachers Service Commission (TSC), Kenya.

RESEARCH AFFILIATIONS

CLARITY-Africa PhD Fellow in Climate Science *Oct 2025 -- Present*
Climate Change Resilient Equitable Healthy Cities in Africa (CLARITY-Africa) | Imperial College London and AIMS-RIC, Kigali

- CLARITY-Africa is a Wellcome-funded project led by Imperial College London with African partners, including AIMS, on the health impacts of climate change in Accra, Tamale, Kigali and Musanze.
- Doctoral research within the programme on flood risk in Rwanda: SAR flood mapping, field validation and flood-depth estimation for Kigali and Musanze.
- Attended the CLARITY-Africa Annual General Meeting (Jun 2026) and presented a PhD progress update.
- *This work was made possible by CLARITY-Africa provided through the AIMS Research and Innovation Centre.*

Member, AXIOM Research Group *Current*
AIMS-RIC, Kigali

- Designed and built the group's website (axiomresearchs.org).

RESEARCH EXPERIENCE

Field campaign for flood validation data *Apr -- May 2026*
Rwanda

- Recorded flood marks, set up measurement equipment and installed sensors at river crossings to validate Sentinel-1 flood extents and calibrate flood-depth estimates.

PRESENTATIONS

GeoAI Africa Community Meeting -- invited talk (online) *24 Sep 2026*
Sentinel-1 SAR flood mapping in Rwanda. A lightning talk had been accepted for the GeoAI for a Climate-Resilient Africa workshop at Deep Learning Indaba 2026 and was moved here.

2nd Rwanda Climate Science Symposium (RCSS 2026) -- oral presentation, Kigali *22--23 Sep 2026*
Session B: Climate Data and Monitoring Systems. Certificate of appreciation.

CLARITY-Africa Annual General Meeting -- PhD progress update *10 Jun 2026*
SAR-based flood mapping for Rwandan catchments (Nyabarongo River Basin).

AIMS-RIC Training School: Data Science Methods -- oral presentation *Mar -- Apr 2026*
SAR-based flood mapping and land-cover classification in Rwanda.

TEACHING AND SUPERVISION

Teaching Assistant (upcoming) *13 -- 27 Jan 2027*
East-African School for Young Researchers on Advanced Machine Learning Techniques (EASY-ML), University of Nairobi, Chiromo Campus

Co-supervisor, MSc research project *2026*
AIMS Rwanda

- Student: Richard Usengimana. Spatial Analysis of Child Malnutrition in Rwanda: District-Level Integration of DHS Data and Agro-Climatic Indicators. Primary supervisor: Dr. Lema Seknewna (AIMS-RIC).

Assistant Lecturer (Part-Time) *Jun 2025 -- Oct 2025*
Vatel Rwanda, Kigali

- Appointed Assistant Lecturer for the 2025--2026 academic year. Delivered ICDL courses to lecturers, staff and students, and supported AI integration in the curriculum.

Physics & Mathematics Teacher *Jan 2021 -- Oct 2021*
Ntanki Secondary School, Meru (Jan -- Apr 2021); Kathiani Boys' High School, Machakos (May -- Oct 2021), Kenya

PROFESSIONAL EXPERIENCE

Head of IT Department (Full-Time) *Jul 2024 -- Oct 2025*
Vatel Rwanda, Kigali

- Led digital transformation of teaching; managed Canvas, Moodle and Triade LMS; built in-house platforms for tracking internships, lecturers, hours and attendance; trained staff and students.

IT Associate -- Mastercard Foundation Associate Enrichment Program *Jul 2023 -- Jul 2024*
Vatel Rwanda, Kigali

- Introduced AI tools in teaching, supported the move to Canvas, and ran training in AI, ICDL and digital literacy.

CONFERENCES, WORKSHOPS & TRAINING

EASY-ML School, University of Nairobi -- teaching assistant (upcoming) *13 -- 27 Jan 2027*

From Satellite Data to Forest Carbon Insights: A Practical GeoAI Bootcamp -- accepted participant (upcoming) *20 -- 22 Oct 2026*
AI.Geolabs, live online.

African Summer School on AI for Weather and Climate Modelling -- accepted participant (upcoming) *19 -- 23 Oct 2026*
ACMAD with Meteo Rwanda and AIMS; AIMS Kigali.

PRESAC-22 Regional Climate Outlook Forum for Central African States -- attended as a doctoral researcher *17 -- 21 Aug 2026*
African Union Commission (ClimSA programme), Kigali.

EarthScope: InSAR Processing and Theory (GMTSAR) *Aug 2026*
Selected from 300+ applicants.

EarthScope Seismology Skill Building Workshop (SSBW) *Jun -- Aug 2026*
70-hour virtual course; Linux & GMT micro-credential.

Geodesy for a Resilient Africa Summer School *Jun -- Jul 2026*
AIMS-Cameroon, Limbe. Built a seismic early-warning web app.

Imperial-AIMS Global Fellows Programme *Jun 2026*
Accra, Ghana. Team lead (PUM-CLID); Best Innovative Accelerator Project.

NASA ARSET: Fundamentals of Remote Sensing -- certificate *Oct 2025*

Ministry of Education, Rwanda: Data Preparation & Analysis Workshop *Sep -- Oct 2024*

ICTP-EAIFR-IUGG Workshop on Computational Geodynamics; 7th ASESMA; Jobberman Finishing School *May -- Jul 2023*

African School of Fundamental Physics and Applications (ASP2022) *Nov -- Dec 2022*
Nelson Mandela University, Gqeberha, South Africa.

African School of Fundamental Physics and Applications (ASP2021) *Jul 2021*
Online.

SELECTED PROJECTS

HeatAlert Accra -- SolarCool CoolTree *Jun 2026*
LSTM-based WBGT heat forecast with CoolShade and CoolBand concepts; dashboard on Hugging Face Spaces. Imperial-AIMS Global Fellows.

TembeaAfrica AI *Mar -- Apr 2026*
Team 5 tool-calling AI travel assistant for Africa (tembeaafrica.github.io). Best Project Award, AIMS-RIC Doctoral Training School, Foundational Methods in Data Science, Kigali.

Seismic Early Warning App *Jun -- Jul 2026*
Seismic data visualisation and alerting (seis-ews-five.vercel.app). Geodesy for a Resilient Africa Summer School.

Object Detection Demo for High Schools *2025*
Real-time item detection for STEM outreach (Google Colab).

AWARDS & HONOURS

Certificate of Appreciation, oral presenter -- 2nd Rwanda Climate Science Symposium *Sep 2026*

Best Innovative Accelerator Project (Team PUM-CLID) -- Imperial-AIMS Global Fellows, Accra *Jun 2026*

Intro Linux & GMT Micro-Credential -- EarthScope / Miami University *Jun 2026*

Best Project Award (TembeaAfrica AI) -- AIMS-RIC Doctoral Training School *2026*

Best 3-Minute Thesis Award -- ICTP-EAIFR PhD & Master's Competition *Jul 2024*

Mastercard Foundation Scholarship -- MSc at AIMS-Senegal *2021 -- 2022*

SKILLS

Programming: Python, R, C/C++, MATLAB, Fortran, Linux shell, LaTeX

Data science & ML: deep learning, machine learning, LSTM, computer vision, statistical learning

Earth observation: SAR/InSAR, optical EO, GPS geodesy, GMTSAR, GMT, remote sensing

IT & EdTech: Canvas, Moodle, web development, ICDL training, Git, Hugging Face Spaces

Languages: English, Swahili, Luhya (fluent)

MEMBERSHIPS

AXIOM Research Group (AIMS-RIC) · Teachers Service Commission (TSC), Kenya · African School of Fundamental Physics and Applications (ASP) · African Strategy for Fundamental and Applied Physics (ASFAP)